Machine Problem Statement

Develop a C program that simulates the deterministic finite automaton (DFA) corresponding to the given regular expression:

$$\text{RE} = 1^*00^*1(0+1)^*$$

Program Requirements

1. **Follow Task #1 (from your Assignment no. 1):**
   - The input string must be a valid string constructed according to Task #1, explicitly applying the operators `*` and `+` as indicated in the instructions.
   - This valid string will serve as the reference for correct DFA simulation.

2. **Flow of the Program:**
   - **Input:** Prompt the user to enter a binary string.
   - **Processing:** Simulate each DFA transition based on the provided regular expression until the end of the string is reached.
   - **Output:** Display one of the following:
     - `"Accepted Input String"` – if the string ends in an accepting state.
     - `"Not Accepted Input String"` – if the string ends in a non-accepting state.

3. **Test Cases:**
   - **Case 1 (Valid):** Input string exactly follows the format from Task #1 (e.g., `"111100001010001111"`). **Expected Output:** `Accepted Input String`.
   - **Case 2 (Invalid):** Modify or omit part of the string so it no longer follows the required format (e.g., `"1100101"` or `"00101"`). **Expected Output:** `Not Accepted Input String`.

This ensures that the DFA correctly identifies and accepts only strings matching the given regular expression, while rejecting those that deviate from the required format.